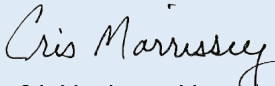


LABORATORY REPORT

Report Number	25-09-120669	Date Submitted	9/16/2025
Material	Fe-Mn-Al-Ni	Date Completed	9/26/2025
Heat	N/A	Reviewed By	 Cris Morrissey – Materials Analyst
Size (in)	N/A		
Submitted By	Song Cai		

Specification: **None referenced**

SUMMARY AND CONCLUSION: **All data collected for informational purposes.**

ADDITIONAL DESCRIPTION

RD0697

REQUESTED ANALYSIS

Chemical analysis

TESTING RESULTS

Interstitial Gas Analysis

Heat	Element/Content (wt%)	
	O	C
RD0697	<0.001	0.010
Test Method	ASTM E1019	

External Chemical Analysis

External Chemical Analysis performed at IMR Test Labs. See attached report 202510308

IMR TEST LABS

A Curtiss-Wright Business Unit
www.imrtest.com

131 Woodsedge Dr.
Lansing, NY 14882
T: 1.607.533.7000 | F: 1.607.533.9210



TEST REPORT **IMR Report Number** **202510308**

September 25, 2025

Material
Fe-Mn-Al-Ni

Cris Morrissey
Fort Wayne Metals
9609 Ardmore Avenue
Fort Wayne, IN 46809
USA

PO Number 2157489

Specification(s)

Date Received September 22, 2025

Quality Provision(s)

Sample(s)

FWM-MTL # 25-09-120669
Size As cast
Reference RD0697

Summary

One sample was received for chemical analysis.

The results are on the following page(s).

All procedures were performed in accordance with the IMR Quality Manual, current revision, and related procedures; and the PWA MCL Manual F-23 and related procedures. The information contained in this test report represents only the material tested and may not be reproduced, except in full, without the written approval of IMR Test Labs ("IMR"). IMR maintains a quality system in compliance with the ISO/IEC 17025 and is accredited by A2LA, certificates #1140.01 and #1140.02. IMR will perform all testing in good faith using the proper procedures, trained personnel, and equipment to accomplish the testing required. Conformance will be based on results without measurement uncertainty applied, unless otherwise requested by the customer. IMR's liability to the customer or any third party is limited at all times to the amount charged for the services provided. All test samples will be retained for a minimum of 3 months and may be destroyed thereafter, unless otherwise specified by the customer. The recording of false, fictitious, or fraudulent statements or entries on this document may be punished as a felony under federal statutes. IMR Test Labs is a GEAE S-400 approved lab (Supplier Code 69805).

Reviewed by

Douglas Miller
Staff Chemist

Chemistry

Element	Sample
Al	8.00
Cr	<0.01
Cu	<0.01
Fe ¹	46.51
Mn	36.49
Mo	<0.01
Ni	8.94
P	0.062
Si	<0.01

¹Determined by difference

Other elements tested (<0.01%): As, B, Be, Bi, Ca, Cd, Co, Mg, Nb, Pb, Sb, Se, Sn, Ta, Ti, V, W, Zn & Zr

Results in weight percent unless otherwise indicated

Method(s): CAP-017U (ICP-AES)